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RSL-NEWS: LARGE-SCALE DATASET AND ENHANCED LIGHTWEIGHT TRANSFORMER MODEL FOR GLOSS-FREE ROMANIAN SIGN LANGUAGE TRANSLATION

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Rezumat

Traducerea Limbajului Semnelor (SLT) a avansat rapid pentru un număr mic de limbi cu resurse bogate, în timp ce Limba Semnelor Române (RSL) a rămas complet în afara acestui domeniu. Singura resursă publică pentru aceasta acoperă semne izolate, ceea ce este suficient pentru recunoaștere, dar face imposibilă traducerea continuă, la nivel de propoziție. Abordăm această problemă cu `RSL-News`, primul corpus RSL la scară largă, fără glose. Prin extragerea a șase ani de știri interpretate de la trei canale naționale de televiziune, am adunat 1.222 de ore de conținut video în limbajul semnelor, aliniat cu vorbirea în limba română transcrisă automat, constând în 648.434 de propoziții și un vocabular de peste 180.000 de cuvinte. Colectarea este posibilă deoarece Legea Audiovizualului din România obligă posturile naționale de televiziune să interpreteze cel puțin 30 de minute de știri zilnice în RSL. Aceeași idee ar putea fi aplicată în orice țară cu o legislație de accesibilitate comparabilă.

Utilizând acest corpus, stabilim primele modele de referință SLT pentru limba română. Pornim de la `SignFormer` [1], un transformer compact fără glose, unde mai întâi identificăm și reparăm două erori ale conexiunilor reziduale care blocaseră convergența pe datele noastre. Apoi abordăm décalage-ul, întârzierea dintre conținutul vorbit și interpretarea sa în limbajul semnelor, extinzând videoclipul dincolo de limitele adnotate ale propoziției și introducând rampe de încredere care reduc importanța cadrelor de care suntem mai puțin siguri că aparțin propoziției curente. În cele din urmă, extragem trăsături de articulare la nivel de fonem cu ajutorul unui model fine-tuned de recunoaștere vizuală a vorbirii și le integrăm end-to-end printr-o conexiune reziduală controlată, evitând etapa intermediară a pseudo-gloselor de care depind sistemele anterioare.

Fiecare componentă aduce un câștig măsurabil, iar modelul final atinge **11.16 BLEU-4** pe setul de testare separat. Acesta este cel mai bun rezultat raportat pe orice set de date cu limbajul semnelor din domeniul transmisiunilor TV, depășind benchmark-urile comparabile `iLSU-T` și `BOBSL`, chiar dacă `RSL-News` are un vocabular mult mai mare. Dincolo de scor, lucrarea introduce limba română în cercetarea SLT și oferă un pipeline reproductibil pentru transformarea interpretărilor obligatorii din transmisiunile TV în date de antrenament pentru alte limbaje ale semnelor cu resurse limitate.

Abstract

Sign Language Translation (SLT) has advanced quickly for a small number of well-resourced languages, while Romanian Sign Language (RSL) has stayed outside the field entirely. The only public resource for it covers isolated signs, which is enough for recognition but makes continuous, sentence-level translation impossible. We address this with `RSL-News`, the first large-scale, gloss-free RSL corpus. By mining six years of interpreted news from three national television channels, we gathered 1,222 hours of signed video aligned with automatically transcribed Romanian speech, consisting of 648,434 sentences and a vocabulary of more than 180,000 words. The collection is possible because the Romanian Audiovisual Law mandates national broadcasters to interpret at least 30 minutes of daily news into RSL. This same idea could carry over to any country with comparable accessibility legislation.

Using this corpus, we establish the first SLT baselines for Romanian. We start from `SignFormer` [1], a compact gloss-free transformer, where we first identify and fix two residual-connection bugs that had been blocking convergence on our data. We then tackle décalage, the lag between spoken content and its signed interpretation, by extending the video past the annotated sentence boundaries and introducing confidence ramps that scale down the frames we are less sure belong to the current sentence. Finally, we extract phoneme-level mouthing features with a fine-tuned visual speech recognition model and integrate them end-to-end through a gated residual connection, avoiding the pseudo-gloss intermediate step that earlier systems depend on.

Every component yields a measurable gain, and the final model reaches **11.16 BLEU-4** on the held-out test split. This is the highest reported result on any broadcast-domain sign language dataset, ahead of the comparable `iLSU-T` and `BOBSL` benchmarks even though `RSL-News` has a much larger vocabulary. Beyond the score, the work brings Romanian into SLT research and offers a reproducible pipeline for turning mandated broadcast interpretation into training data for other under-resourced sign languages.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Problem Context and Significance

Sign languages are the primary means of communication for tens of millions of Deaf and Hard of Hearing people worldwide [2]. They are not pantomime or a visual rendering of speech, but fully formed natural languages, each with its own grammar, lexicon, and regional variation [3]. Despite this, communication between Deaf and hearing people in everyday situations has barely improved over the decades; the usual workarounds, passing handwritten notes or typing back and forth on a phone, are slow and break the natural flow of a conversation. Automatic Sign Language Translation (SLT), turning a video of someone signing into written text in the surrounding spoken language, is the technology that could remove this barrier, yet it remains far behind its speech and text counterparts.

Part of the reason lies in how the field has traditionally been built. For most of its history, SLT relied on glosses, written labels that transcribe each sign one by one, as an intermediate representation between video and text. Glossing is accurate but expensive: it has to be done by hand by trained linguists, and annotating a minute and a half of video takes roughly an hour [4]. This cost caps glossed datasets at a few hundred hours at most, and it discards everything that is not a manual sign, including the facial expressions and mouthings that carry grammatical meaning. Recent work has therefore moved toward gloss-free, end-to-end translation, learning the mapping straight from video features to text [5]. Removing the annotation step is what finally lets these systems scale to the thousands of hours of signing that already sit unused in broadcast archives and online video.

This scaling, however, has mostly benefited a small set of well-resourced languages; American, German, and Chinese Sign Language attract the bulk of the datasets and the models [6]–[8]. Romanian Sign Language is absent from this picture. It is the native language of tens of thousands of people in Romania, yet the only public resource for it covers isolated signs [9], which suffices for dictionary-style recognition but not for translating connected, sentence-level signing. There is no continuous RSL corpus, and consequently no published translation results to build on.

There is, on the other hand, a steady and almost untapped source of data. The Romanian Audiovisual Law No. 504/2002 requires national broadcasters to interpret at least 30 minutes of news or debate programs into RSL every day [10]. Six years of these broadcasts amount to a large, naturally aligned collection of signing paired with the spoken content it translates, all of it public. This work is built on that observation. We mine three national channels to construct `RSL-News`, the first large-scale gloss-free RSL dataset, and then use it to establish the first SLT baselines for Romanian, extending a compact translation model with techniques aimed at the specific difficulties of interpreted broadcast data.

1.2 Objectives

In this work, we aim to:

1. Build the first large-scale, continuous RSL corpus by mining public news broadcasts, together with a semi-automated pipeline (download, interpreter cropping, temporal segmentation, transcription, and alignment) that can be reused for other sign languages with similar broadcast sources.
2. Establish the first SLT baselines for Romanian on this corpus, using `SignFormer` [1], a small gloss-free transformer, as the starting point.
3. Model the interpreter delay (décalage) between speech and signing, which causes signed content to flow across the sentence boundaries we derive from the audio.
4. Extract phoneme-level mouthing features and integrate them end-to-end into the translation model, testing whether the words interpreters silently mouth help disambiguate visually similar signs.

1.3 Scope and Limitations

`RSL-News` is a news corpus, so the register is formal and the topics track whatever was in the headlines that day; a model trained on it will not handle casual conversational signing well without further adaptation. The footage is real broadcast material from three channels and six interpreters, not studio recordings, which keeps conditions realistic but narrow, and it is far from balanced: one signer alone accounts for most of `Digi24` and `PrimaTV`. Translation goes one way only, from signed video to Romanian text. We never produce glosses at any point in the pipeline. The supervision is itself a limitation, since it comes from automatic transcription rather than human transcripts, so the targets carry some ASR noise; and because interpreters compress and reword as they go, what the model learns to output is their rendering of the news rather than a literal copy of the anchor. Every result we report is a BLEU-4 score.

1.4 Structure of the Work

Chapter 2 sets up the rest: the sign language understanding tasks, why glosses became a bottleneck, and the gloss-free transformers and feature extractors we build on, ending with the gaps this work targets. The dataset is the subject of Chapter 3, which goes from where the data comes from and the law that makes it available, through the collection and alignment pipeline, to the final statistics and a comparison against existing corpora. Chapter 4 turns to the modeling: the two `SignFormer` bugs we tracked down, the strategies for handling décalage, including the confidence ramps, and how the mouthing features are extracted and fused in. Chapter 5 reports the experiments, ablating each component and placing the final number next to other broadcast-domain systems. Chapter 6 closes the work and lays out what is left to do.

Chapter 2

Background and Related Work

2.1 Sign Language Understanding Tasks

Sign Language Understanding (SLU) sits at the intersection of computer vision (CV) and natural language processing (NLP). It is comprised of three distinct, progressively complex tasks:

Isolated Sign Language Recognition (ISLR) treats recognition purely as classification: given a short, pre-segmented video clip with a single sign, ISLR predicts the corresponding sign label. It is effectively a classification problem and cannot handle natural (continuous) signing [9] (because it operates with known temporal boundaries and does not have the complicated co-articulations between consecutive signs).

Continuous Sign Language Recognition (CSLR) extends this to sequences. Given an unsegmented video of continuous signing, CSLR produces an ordered sequence of glosses (gloss = written label representing an individual sign). CSLR is typically trained with Connectionist Temporal Classification (CTC) loss, which handles the dynamic alignment between input frames and output glosses. The main challenge is the co-articulation; the boundaries between consecutive signs are blurred by the transition motions between signs. Furthermore, these transitions vary depending on neighboring signs.

Sign Language Translation (SLT) is the most complex task. Rather than producing glosses, SLT generates natural language text directly from videos of continuous signing. This requires understanding not only the lexical content but also the grammar and syntax of both the sign language and the spoken language. This is challenging because the two languages often differ substantially in word order and morphology [7].

Historically, SLT was performed in two stages; first, a CSLR system transcribes video into glosses, and then a neural network translates the gloss sequence into natural language. While this pipeline achieves strong results on small benchmark datasets, it is heavily bottlenecked due to gloss representation. The alternative, direct end-to-end SLT from visual features to text, avoids this bottleneck and is the paradigm we will use.

2.2 The Gloss Bottleneck and Gloss-Free Approaches

A gloss is a written annotation of a sign, typically a word from the surrounding spoken language used as a label (e.g., `HOUSE`, `TOMORROW`). Gloss annotation requires trained linguists familiar with the specific sign language. It is labor-intensive and takes on average one hour per 90 seconds of video [4]. This dependency creates what we term the *gloss bottleneck*, which manifests in several ways:

1. **Dataset size limitation:** The cost of expert annotation restricts glossed datasets to at most a few hundred hours. The largest fully-glossed dataset [11] reaches approximately 100 hours; by contrast, gloss-free corpora exceed 1,000 hours.
2. **Loss of non-manual markers:** Glosses encode only the manual component of signing. Facial expressions, mouthing, eye gaze, and body posture, all of which carry grammatical and semantic information, are discarded.
3. **Inability to use unlabeled data:** Vast quantities of sign language video exist in broadcast archives and online platforms, but they cannot be used by gloss-supervised pipelines without annotation.

Therefore, recent research has shifted towards gloss-free approaches, eliminating the need for gloss annotation and learning a direct mapping from sign language videos to text [5]. This paradigm shift enables scaling to orders of magnitude more data: `BOBSL` contains 1,467 hours of BSL [12], `YouTube-ASL` provides 984 hours of ASL [6], `CSL-News` reaches 1,985 hours of Chinese Sign Language [8], and our `RSL-News` contributes 1,222 hours of Romanian Sign Language.

Recently, gloss-free methods have reached gloss-supervised levels of accuracy. On `CSL-Daily` (a controlled lab dataset with 2K vocabulary), `Uni-Sign` achieves 27.78 BLEU-4 without gloss supervision [8], very close to the best gloss-based systems. The evolution of gloss-free methods is as follows: Camgöz et al. [13] first used the encoder-decoder transformer in 2020; `GASLT` [14] introduced temporal locality biases in 2023 to model gloss supervision in gloss-free datasets; `Sign2GPT` [15] incorporated pretrained LLMs in 2024; `SignFormer` [1] demonstrated that the carefully tuned architecture can match the raw scale that same year; and `Uni-Sign` showed in 2025 that unified training on thousands of hours can match gloss-supervised performance.

2.3 Transformer Architectures for Gloss-Free SLT

2.3.1 Joint Sign Language Transformers

Camgöz et al. [13] introduced the first transformer-based architecture for SLT. The model jointly trains sign language recognition (via CTC loss on encoder outputs) and translation (via cross-entropy on decoder outputs), with recognition loss weighted $5\times$ to encourage temporally structured encoder representations.

On `PHOENIX-2014T` (a wild, single-domain weather dataset with 9 signers and 3K vocabulary), this achieved 21.80 BLEU-4, more than double the prior state of the art of 9.58

BLEU-4. It also surpassed the Gloss2Text performance of 19.26, indicating that the joint model captures information beyond what glosses alone encode. Although the system still requires gloss supervision for the CTC auxiliary loss, its multi-head cross-attention architecture became the foundation for subsequent SLT transformers.

2.3.2 GASLT

The key insight of `GASLT` (Gloss Attention for Sign Language Translation) [14] is that gloss supervision provides two properties to the encoder: (1) temporal localization, causing each output token to attend primarily to a local window of input frames, and (2) sentence-level semantic alignment. `GASLT` replicates both properties without glosses.

For temporal localization, the model introduces *Gloss Attention*: each attention head learns dynamic offsets that restrict its attention to a temporally local range. This produces the diagonal pattern in attention maps (where output tokens attend sequentially along the input timeline) without explicit gloss boundaries. For sentence-level understanding, `GASLT` transfers knowledge from `Sentence-BERT` [16] by training the encoder's pooled output to match the sentence embedding of the target text.

`GASLT`'s Gloss Attention module demonstrated that architectural constraints can substitute expensive annotation.

2.3.3 SignFormer

`SignFormer` [1] enhances `GASLT` with two additions: a 1D convolution module and dual position encodings. The convolution module follows a pointwise-depthwise-pointwise factorization inserted after each self-attention layer. This captures local temporal patterns that attention alone may miss. For position encoding, the model combines Absolute Position Encodings (`APE`) for global structure with Context-aware Position Encodings (`CoPE`) [17]. This conditions positional information on the local context and adapts the notion of distance to the signing speed.

With only 3.88M parameters, `SignFormer` achieves 23.43 BLEU-4 on `PHOENIX-2014T` in the gloss-free setting, competitive with models exceeding 1 billion parameters. This result demonstrates that careful architectural design can substitute for raw scale. We adopt `SignFormer` as the baseline architecture for all translation experiments. The bugs we discovered and fixed, along with our extensions for modeling interpreter delay and integrating mouthing features, are detailed in Chapter 4.

2.3.4 Sign2GPT

`Sign2GPT` [15] uses a pretrained 1.7B parameter LLM as the decoder. To bridge the modality gap, the authors introduce *pseudo-gloss pretraining*: target sentences are processed with `spaCy` lemmatization to extract pseudo-glosses, `fastText` for embeddings, and the sign encoder is trained to align its frame-level outputs with these embeddings using cosine similarity.

The visual encoder uses DINOv2 [18] with LoRA [19], keeping the pretrained weights largely frozen. Cross-attention layers between encoder and LLM decoder are initialized with zero-gating. On PHOENIX-2014T, Sign2GPT achieves 23.20 BLEU-4 in the gloss-free setting, demonstrating that it is possible to transfer linguistic knowledge from pretrained LLMs to SLT.

2.3.5 Uni-Sign

Uni-Sign unifies the 3 tasks; ISLR, CSLR, and SLT, under the same pretraining process, and treats each as a downstream fine-tuning task. The model processes 69 whole-body keypoints through a Spatial-Temporal Graph Convolutional Network (ST-GCN) encoder [20], (optionally) fuses pose and RGB streams, and decodes with mT5-Base [21].

The authors also introduce CSL-News, a 1,985-hour gloss-free Chinese Sign Language dataset from TV broadcasts used for large-scale pretraining. The system achieves 26.86 BLEU-4 on CSL-Daily and 14.9 BLEU-4 on How2Sign (controlled lab, 16K vocabulary, instructional domain), demonstrating that large-scale data and unified training objectives can match gloss-supervised performance. This motivated our own large-scale data collection for Romanian Sign Language.

2.3.6 SignBind-LLM

SignBind-LLM [22] addresses two components of signing that current approaches do not capture: fingerspelling and mouth cues. The architecture consists of three separate expert predictors (continuous signing, fingerspelling, and lipreading), each pretrained independently. A lightweight fusion transformer with a learnable gating mechanism temporally synchronizes these streams before combining and passing them to a LLaMA-7B decoder [23].

The lipreading expert extracts phoneme-level features from the signer's mouth region via a ViT encoder with CTC decoding. Their study confirms that mouthing features provide a strong complementary signal: removing lipreading drops BLEU-4 from 22.1 to 13.6 on How2Sign.

The key difference in our approach is the decoding paradigm. SignBind-LLM uses its expert predictors to generate pseudo-glosses which are then processed (reordered and filled in) by the LLM. Our approach is end-to-end: we inject phoneme features directly into the SignFormer decoder via a sigmoid gate, without any intermediate pseudo-gloss step. Their system achieves 22.1 BLEU-4 on How2Sign and 6.8 on BOBSL (wild, diverse BBC content, 77K vocabulary); we achieve 11.16 on RSL-News (wild, news domain, 180K vocabulary) test split.

2.4 Feature Extraction for SLT

Sign language translation systems do not operate on raw pixels; instead, they usually use pretrained feature extractors that convert video frames into numerical representations. The two main paradigms are skeleton-based pose estimation and learned RGB features.

2.4.1 Pose Features

Pose estimation extracts body, hands, and face landmarks as coordinates from each frame. The resulting skeleton representations are invariant to noise (robust to clothing, lighting, and background changes) and very compact; however, they lose fine-grained visual details such as hand shape textures and subtle finger configurations.

In this work, we use `RTMW-x` [24], which estimates 133 whole-body keypoints (17 body, 42 hands, 68 face, 6 feet).

2.4.2 RGB Features

The learned RGB feature extractors encode appearance and motions directly from pixels.

For sign language, Albanie et al. [25] fine-tuned `I3D` (Inflated 3D ConvNet) [26] on `BSL-1K`, a 1,000-hour BSL broadcast dataset with 1,000 sign classes obtained through mouthing-based weak supervision: subtitles are aligned to video via keyword spotting on automatically detected lip movements, providing training labels without manual annotation. Varol et al. [27] extended this to `BSL-5K` with approximately 5,000 classes. The resulting model has become a standard feature extractor for SLT and is our extractor of choice.

The extraction pipeline processes 8-frame windows with stride 2 through the network up to the `Mixed_5c` layer, then spatially mean pools to produce 1024-d feature vectors.

2.4.3 Mouthing and Visual Speech Features

Sign language interpreters frequently mouth words from the spoken language while signing. This behavior has been used for dataset construction (`BSL-1K` uses automatic lip-reading to generate weak supervision) and recently for translation itself via `SignBind-LLM`.

`Auto-AVSR` [28] represents the state of the art in audio-visual speech recognition. It combines a `ResNet-18` [29] visual frontend with a `Conformer` [30] encoder and a hybrid CTC/attention decoder [31]. Through self-supervised pretraining and pseudo-label-based self-training, it achieves a 0.9% word error rate on `LRS3` [32] in video-only mode. The hybrid CTC/attention architecture jointly optimizes CTC loss (enforcing monotonic alignment) and attention-based sequence-to-sequence loss (modeling language and flexible output dependencies).

In this work, we fine-tune `Auto-AVSR` on Romanian speech data with phonemes as targets obtained via `RoLEX` [33], then inject the resulting features into the `SignFormer` decoder through a learned sigmoid gate. Unlike `SignBind-LLM`, which feeds lipreading outputs through an LLM for pseudo-gloss reordering, our integration is end-to-end within the translation model. Details are given in Chapter 4.

2.5 Existing SLT Datasets

Data scarcity is an issue in SLT research. Early work relied mostly on `PHOENIX-2014T`; only recently have larger datasets, gloss-free datasets, become available. Table 2.1 provides a quantitative overview; we include domain information to contextualize BLEU scores, since controlled lab recordings and single-domain datasets are easier to learn than wild broadcast content with diverse topics.

Table 2.1: Comparison of SLT datasets. Best BLEU-4 scores from [5]. “controlled” = lab-recorded; “wild” = broadcast/web.

Dataset	Language	Hours	Signers	Vocab	Setting	Domain	Best BLEU-4
CSL-Daily [8], [34]	CSL	23	10	2K	Controlled	Daily	26.86
How2Sign [4]	ASL	79	11	16K	Controlled	Instructional	15.5
PHOENIX-2014T [7]	DGS	11	9	3K	Wild	Weather	26.75
iLSU-T [35]	LSU	202	18	38K	Wild	News	3.43
OpenASL [36]	ASL	288	220	33K	Wild	Open	13.21
YouTube-ASL [6]	ASL	984	2500+	60K	Wild	Open	–
BOBSL [12]	BSL	1,467	39	77K	Wild	Diverse	7.3
CSL-News [8]	CSL	1,985	–	5K	Wild	News	–
RSL-News (ours)	**RSL**	**1,222**	**6**	**180K**	**Wild**	**News**	**11.16**

It can be noticed that BLEU-4 scores are strongly correlated with vocabulary size, domain diversity, and recording conditions. Narrow-domain datasets (`PHOENIX-2014T` with weather vocabulary) yield high scores (26.75); wild diverse-domain datasets with large vocabularies yield much lower scores (`BOBSL` at 7.3 with 77K vocabulary; `iLSU-T` at 3.43 with 38K vocabulary). Our 11.16 BLEU-4 on `RSL-News` (180K vocabulary, wild news domain) test split is consistent with this and represents a strong result among large-scale broadcast datasets.

2.6 Gap Analysis

This work aims to address several gaps identified in the current SLT research landscape:

No continuous RSL data. `RoCoISLR` [9] provides only isolated signs. No dataset supports continuous RSL recognition or translation, leaving Romanian absent from the SLT research landscape.

No SLT baselines for Romanian. As a direct consequence, there exist zero published results on RSL-to-text translation. Our work establishes the first such baselines.

Interpreter delay lacks a universal solution. The temporal offset between spoken content and its signed interpretation (*décalage*) causes information to leak across sentence boundaries. Current solutions are limited: `BOBSL` relies on a complex temporal alignment model incompatible with our dataset, while `iLSU-T` uses basic random delay augmentations. We evaluate alternative modeling techniques in Chapter 4 and propose an approach that outperforms the `iLSU-T` baseline on our data.

Mouthing is underexplored in SLT. `SignBind-LLM` recently demonstrated that lip movement features improve translation quality, but their approach routes lipreading through a pseudo-gloss intermediate representation. No prior system integrates phoneme-level mouthing features directly into an end-to-end translation decoder.

To address these gaps, we present `RSL-News`, a large-scale RSL dataset, fix the `SignFormer` architecture, extend it with décalage modeling, and integrate mouthing features end-to-end into the translation decoder.

Chapter 3

The RSL-News Dataset

Romanian Sign Language lacks any continuous, sentence-level datasets required for training translation models. Existing resources are limited to isolated sign recognition [9], making end-to-end SLT impossible. To address this gap, we constructed RSL-News. It is the first large-scale gloss-free RSL corpus, consisting of 1,222 hours of interpreted news broadcasts aligned with automatically transcribed Romanian speech. The dataset is extracted from three national TV channels and contains six years of daily news programs.

3.1 Data Sources and Legal Context

The Romanian Audiovisual Law No. 504/2002, Article 42[1] [10], requires that national TV broadcasts contain at least 30 minutes of news or debate programs interpreted into RSL, per day. This legal requirement creates a natural and growing source of aligned sign-speech data: interpreters translate the spoken content of news anchors into RSL in real time. This produces a strong, natural temporal correspondence between the audio stream and the signed video that can be used for SLT without explicit gloss annotation.

We selected three TV channels that respect this legal requirement and offer publicly accessible archives:

1. **Digi24**, a 24-hour news channel. The 11:00 AM News Journal features a sign language interpreter. Content was obtained from the public video archive at `digi24.ro`.
2. **ProTV**, Romania's largest national broadcaster. The Morning News program includes RSL interpretation. Content was scraped from the public archive at `stirileprotv.ro`.
3. **PrimaTV**, a national channel whose `FOCUS` news program features RSL interpretation. Content was downloaded from the station's official `YouTube` channel.

The collected data are from November 2019 to November 2025, 2,895 episodes in total. `Digi24` and `ProTV` together comprise the bulk of the dataset in both episode count and total duration.

Table 3.1: Data source summary.

Channel	Date Range	Source	Episodes
Digi24	2019-11-12 – 2025-07-04	Website archive	1,200
ProTV	2021-01-20 – 2025-11-08	Website archive	1,117
PrimaTV	up to 2025-11-15	YouTube	578
Total			**2,895**

3.2 Collection Pipeline

Broadcast episodes are usually multiple hours long and contain only approximately 30 minutes of sign language-interpreted footage, constrained to a small on-screen region. Extracting data suitable for SLT training required a multi-stage semi-automated pipeline: video download, interpreter region-of-interest (ROI) cropping, and temporal segmentation to isolate only the portions where the interpreter is actively signing.

3.2.1 Video Download

Each channel required a distinct download strategy due to differences in archive structure and access patterns.

Digi24. Episodes were scraped from the `digi24.ro/emisiuni/jurnale` video archive. The scraper navigated the pages of the listed categories to collect episode page URLs. Because the video players on the site are JavaScript-based and do not expose direct video URLs, we used `Selenium` to interact with each page's player, select the highest available quality, and extract the underlying video source URL. All videos were downloaded at their native resolution of 1280×720 pixels, 25 fps.

ProTV. The archive at `stirileprotv.ro/video/stirile-diminetii/` organizes episodes by date in the page title. We reconstructed direct video URLs from the title dates; when the initial URL returned a 404 error (due to date mismatches between the title and the actual upload date), a binary search within a ± 2-year window located the correct URL.

PrimaTV. Episodes were downloaded from public `YouTube` playlists. We used `yt-dlp` configured to download the best available video stream and the Romanian audio track. Because the playlist contained non-FOCUS content (thus contained no interpreters), we filtered episodes based on title keywords and performed manual thumbnail verification to confirm relevance.

3.2.2 Interpreter Region of Interest

For all three channels, the sign language interpreter is placed in a fixed rectangular region, in the bottom-right corner of the video. However, the exact coordinates of this box vary across time periods as channel graphics change. We determined the coordinates by manually inspecting the videos.

Digi24 underwent a format change in late 2023. For broadcasts from 2024 onward (and December 2023), the interpreter box is located at $x \in [966, 1216]$, $y \in [287, 510]$, yielding a 250×223 pixel crop. For earlier broadcasts, the region shifts to $x \in [1018, 1210]$, $y \in [474, 680]$ (192×206 pixels).

ProTV episodes appear at two resolutions: at a height of 720p, the crop is $x = 952$, $y = 347$, 300×322 pixels; at a height of 432p, the crop is $x = 571$, $y = 208$, 180×193 pixels. The pipeline detects height automatically and applies the corresponding coordinates.

PrimaTV episodes appear at two resolutions: at 1080p the crop is $x = 1522$, $y = 720$, 383×411 pixels; at 720p the crop is $x = 1015$, $y = 480$, 255×274 pixels. The pipeline detects resolution automatically and applies the corresponding coordinates.

3.2.3 Temporal Segmentation

A broadcast is not only an interpreted footage. Commercial breaks, foreign-language reports, the gaps between news items; none of these carry an interpreter, and all of them have to be removed. What we keep are the segments where the interpreter is on screen and actively signing, and since the three channels lay out their programs differently, there are some differences in their segmentation strategies.

3.2.3.1 Digi24: Multi-Segment Extraction

`Digi24` is the hard case. While `ProTV` and `PrimaTV` contain a single contiguous interpreted block of around 30 minutes, `Digi24` scatters its interpretation across the program in short bursts, the median one being under 7 minutes long. Finding all of them needs a three-stage pipeline: a pose-based heuristic, a binary classifier trained on top of it, and a state machine that stitches the frame-level predictions back into segments.

Step 1: Pose Heuristic. We sample 40 frames uniformly from each video and extract keypoint skeletons using `rtmlib Wholebody` [37]. A frame is classified as containing an active interpreter if and only if: (a) exactly one person is detected; (b) the person faces the camera, verified by shoulder symmetry (the absolute difference between left and right shoulder y-coordinates must not exceed 20% of the shoulder span); (c) both elbows and wrists are detected with confidence ≥ 0.5; and (d) the framing is waist-up, consistent with the interpreter box layout. The pose model used here is fast but not very accurate, so these labels are noisy, good enough to bootstrap the classifier in the next step but not to trust on their own.

Step 2: Binary Classifier. For the classifier itself, we freeze a `MobileNetV3 Small` [38] and train only a classification head on top, separating frames with an interpreter from frames without an interpreter. Inputs are normalized with ImageNet statistics, and we lean on augmentation to handle the class imbalance. Training stops as soon as the validation accuracy crosses 99.5%. The result is light enough to run in real time on a single consumer

GPU, which is what made it feasible to push every frame of every `Digi24` episode through it.

Step 3: Stateful Segmentation. The classifier produces a verdict per frame; a state machine then walks that sequence and groups the frames into segments, governed by four thresholds:

- *Disappearance tolerance*: 2 seconds. Brief absences, such as graphic overlays momentarily occluding the interpreter, are ignored.
- *Long absence*: 90 seconds. Interpreter absence beyond this threshold ends the current episode search.
- *Minimum segment duration*: 10 seconds. Segments shorter than this are discarded as noisy detections.
- *Presence confirmation*: 2 seconds. The interpreter must be continuously detected for this duration before a new segment begins.

Step 4: Segment Extraction. The temporal and spatial boundaries are passed to `FFmpeg` to crop the interpreter box and extract the audio track for each segment.

3.2.3.2 ProTV and PrimaTV: Contiguous Block Detection

`ProTV` and `PrimaTV` broadcasts usually contain a single contiguous block of interpreted footage of approximately 25–30 minutes. Because only one segment needs to be located and extracted per episode, we use a simpler face recognition-based approach.

Step 1: Face Recognition. We pre-compute face encodings for all interpreters using the `face_recognition` library [39].

Step 2: Strided Search. The pipeline samples frames at 15-minute intervals across the video duration. For `PrimaTV`, the stride is adaptive: $\min(15\,\text{min}, \text{frame\_count}/4)$ to handle shorter episodes. For each sample, the frame is checked against the interpreter face encodings. Once a match is found, the approximate location of the interpreted block is known.

Step 3: Binary Search. Starting from the matched position, a binary search finds the exact start and end timestamps of the interpreted block.

Step 4: Verification. After boundary detection, the pipeline searches 10 minutes before and after the identified block using the `MobileNetV3` classifier to detect any missed content or stray interpreted segments.

Step 5: Validation. Segments shorter than 10 minutes are rejected, as they likely represent detection errors given the legal minimum of 30 minutes of daily interpreted content.

Across all channels, these 2 strategies produced **4,632 contiguous segments** of sign language content.

3.3 Transcription, Alignment, and Filtering

The sign language interpreter translates the broadcast speech content; therefore, we can derive textual supervision from the spoken audio for SLT training. We extract sentence-level transcriptions aligned through timestamps by filtering foreign languages, transcribing the speech, aligning the transcriptions, and doing post-processing.

3.3.1 Language Filtering

Romanian news broadcasts occasionally include foreign-language segments (interviews, press conferences, dubbed reports). Since our translation targets are Romanian sentences, we filter out non-Romanian speech prior to transcription.

Audio is extracted from each segment at 16 kHz. `Silero-VAD` [40] chunks the audio into speech segments using a silence threshold of 1 second. Each chunk is then passed through the `WhisperX` `large-v3-turbo` encoder [41] followed by a language detection head. The result is a binary mask: `True` where Romanian is detected, `False` otherwise. Masks are stored as packed bits for space efficiency.

3.3.2 Transcription

Non-Romanian audio regions are zeroed out using the language mask, thus the transcription model will only process Romanian speech. The modified audio is then transcribed by `WhisperX` using the `large-v3` model with the language explicitly set to Romanian. We use the full `large-v3` model (rather than the turbo variant) for transcription quality; the turbo model is only used as the fast language detection pass, where speed matters more than accuracy.

3.3.3 Sentence-Level Alignment

Raw transcriptions provide only coarse timestamps (20–30 second chunks). For SLT training, we need sentence-level mappings for the video frame indices. We use the `WhisperX` alignment model (`whisperx-align_ro`), which provides word-level timestamps. The alignment model internally handles sentence boundary detection and produces sentence-level start and end timestamps that we can use directly to map transcriptions to video segments.

3.3.4 Post-Processing and Filtering

ASR introduces artifacts and noise that must be removed before the text can be used as the training target. We apply the following filters to each sentence:

Character Normalization. Romanian text generated by `WhisperX` occasionally uses cedilla diacritics instead of the standard comma-below diacritics. We replace all instances.

Duration Filters. Sentences shorter than 2 seconds are discarded (too brief to contain meaningful signed content). Sentences where the average word duration falls below 0.1 seconds or exceeds 2 seconds are also discarded, as these indicate alignment errors (impossibly fast articulation or broken timestamps, respectively).

Hallucination Detection. `WhisperX` is known to enter repetition loops under certain audio conditions (noisy background, silence). We scan each transcription for consecutive repetitions of any phrase of length between 1 and 64 words. If a repeated phrase occurs ≥ 3 times consecutively and the repetition dominates $\geq 50\%$ of the total word count for that sentence, the sentence is discarded. This filter catches hallucinations without removing the actual repeated content.

Lowercase Normalization. For the following experiments, all text is lowercased. However, a case-preserving variant is also maintained.

3.3.5 Tokenization

We train a `SentencePiece Unigram` language model [42] on the full filtered dataset. The vocabulary size is set to 16,384 subword tokens with 100% character coverage. Four special tokens are reserved: <unk> (ID 0), <pad> (ID 1), <s> (ID 2), and </s> (ID 3). We produce 2 tokenizer variants: one trained on lowercase text and one on case-preserving text.

With over 180,000 unique forms, 51.7% of which are singletons, a word-level vocabulary would be impractically large and sparse. Subword tokenization allows the model to decompose rare inflected forms into shared roots.

3.4 Feature Extraction Overview

Visual feature extraction transforms video frames into numerical representations for the translation model. We derive three complementary feature streams. The technical details and architectural choices for each extractor are discussed in Chapter 4.

RGB Features. Frames are resized so that the larger side is 256 pixels, then center-cropped to 224×224 and processed in temporal windows of 8 frames with stride 2 through an `I3D` network pretrained on `BSL-5K`. The output is a sequence of 1024-d feature vectors.

Pose Features. Each frame is scaled and letterboxed to 192×256 and processed by `RTMW-x`, giving us 133 whole-body keypoints (17 body, 42 hands, 68 face, 6 feet) with confidence scores. An example visualization is shown in Figure 3.1.

Mouthing Features. The mouth region is cropped and processed by a fine-tuned `Auto-AVSR` model to extract phoneme-level features that capture the interpreter's mouthings.

All features are pre-extracted and stored offline. During training, sentence timestamps are converted to frame indices, allowing efficient random access using memory-mapping.

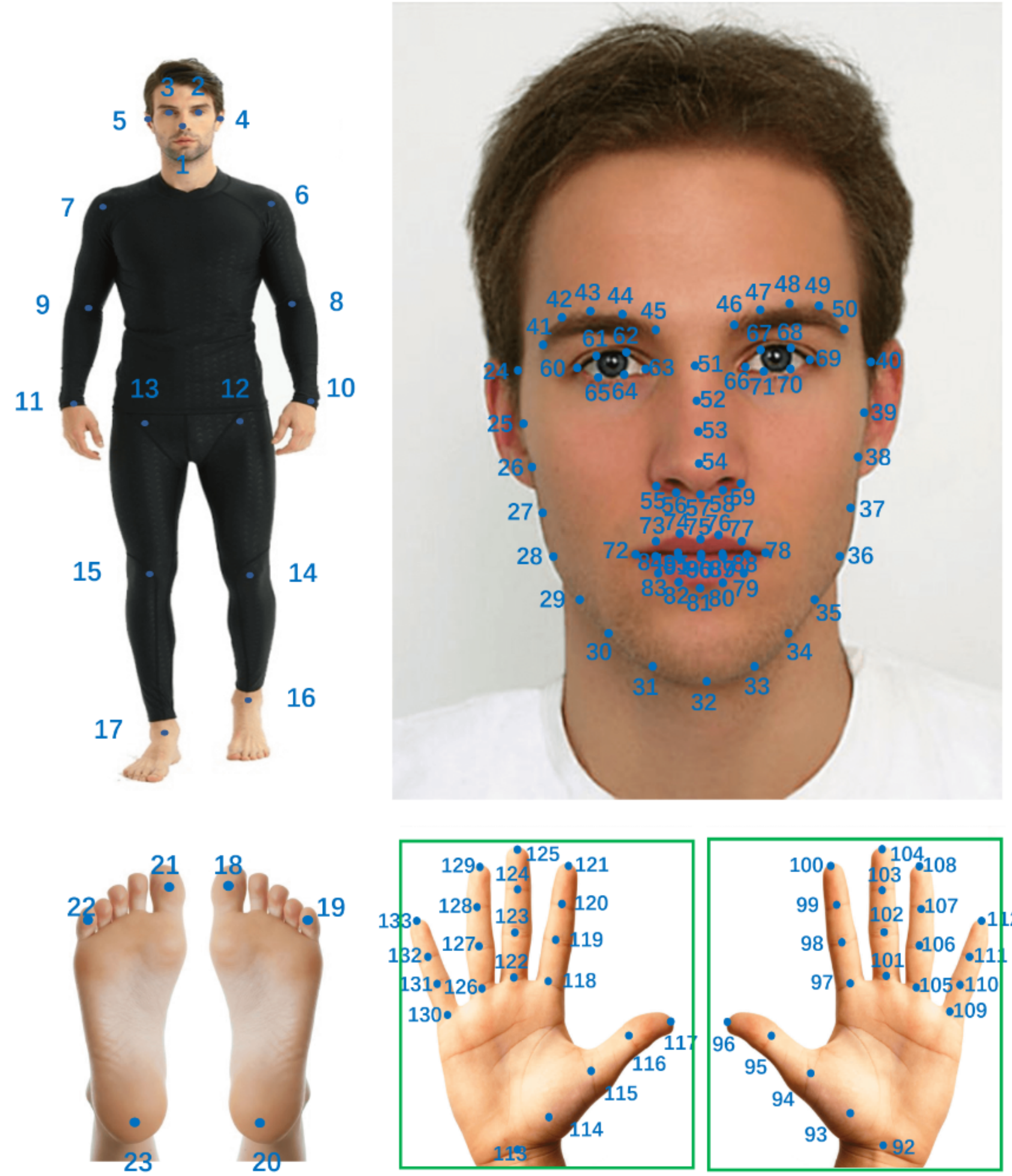

Figure 3.1: Visualization of whole-body pose keypoints extracted by RTMW-x, following the COCO-WholeBody format. [43]

3.5 Dataset Statistics

`RSL-News` consists of **1,222 hours and 39 minutes** of signed video, split across 2,895 episodes and 4,632 contiguous segments. The segments contain a total of 648,434 sentences, and the text consists of 11,155,599 words drawn from a vocabulary of 180,892 unique forms. Table 3.2 gives the per-channel breakdown.

Table 3.2: Per-channel statistics of the `RSL-News` dataset. "Singletons" are vocabulary items appearing exactly once. Lemma counts are computed using `spaCy` with `Stanza` [44] as the underlying NLP pipeline.

Channel	Ep.	Seg.	Sent.	Duration	Vocab	Lemmas	Singletons	% Single	Total Words	Avg Dur	Avg Len
PrimaTV	578	578	145,292	249h 45m	82,505	48,349	41,515	50.3%	2,370,825	6.19s	16.3w
Digi24	1,200	2,813	222,082	459h 02m	92,965	55,838	46,014	49.5%	4,172,034	7.44s	18.8w
ProTV	1,117	1,241	281,060	513h 51m	114,893	69,442	56,406	49.1%	4,612,740	6.58s	16.4w
Total	**2,895**	**4,632**	**648,434**	**1222h 39m**	**180,892**	**111,974**	**93,547**	**51.7%**	**11,155,599**	**6.79s**	**17.2w**

`Digi24`'s sentences run longer on average, 18.8 words against roughly 16 for `ProTV` and `PrimaTV`. This likely reflects the more formal, exposition-heavy style of `Digi24`'s midday journal versus the conversational morning format of `ProTV` and the shorter news segments of `PrimaTV`'s `FOCUS` program. Duration follows the same split, 7.44 seconds per sentence on `Digi24` against 6.19–6.58 on the other two.

At 51.7%, the singleton rate looks high until we account for how Romanian works. It is a heavily inflected language with noun declensions, verb conjugations, and adjective agreements, and a single root can surface as dozens of distinct forms depending on case, number, gender, tense, and mood. Even in a corpus this large, that leaves a long tail of forms seen exactly once. Lemmatizing with `spaCy` and `Stanza` confirms this: the 180,892 surface forms collapse to 111,974 lemmas, a 38% drop that pins most of the singletons on morphology rather than on truly rare words. The subword tokenizer from Section 3.3.5 handles the rest, splitting uncommon forms into pieces that it has already seen.

Figure 3.2 plots segment duration by channel. `ProTV` and `PrimaTV` sit around the 25–30 minute mark, each segment being essentially one full interpreted block, whereas `Digi24` ranges from under a minute to over half an hour, with the fragmented structure from Section 3.2.3.1 showing through. Sentence durations and lengths, in Figure 3.3 and Figure 3.4, come out as right-skewed, the usual shape for natural text where very long sentences are rare.

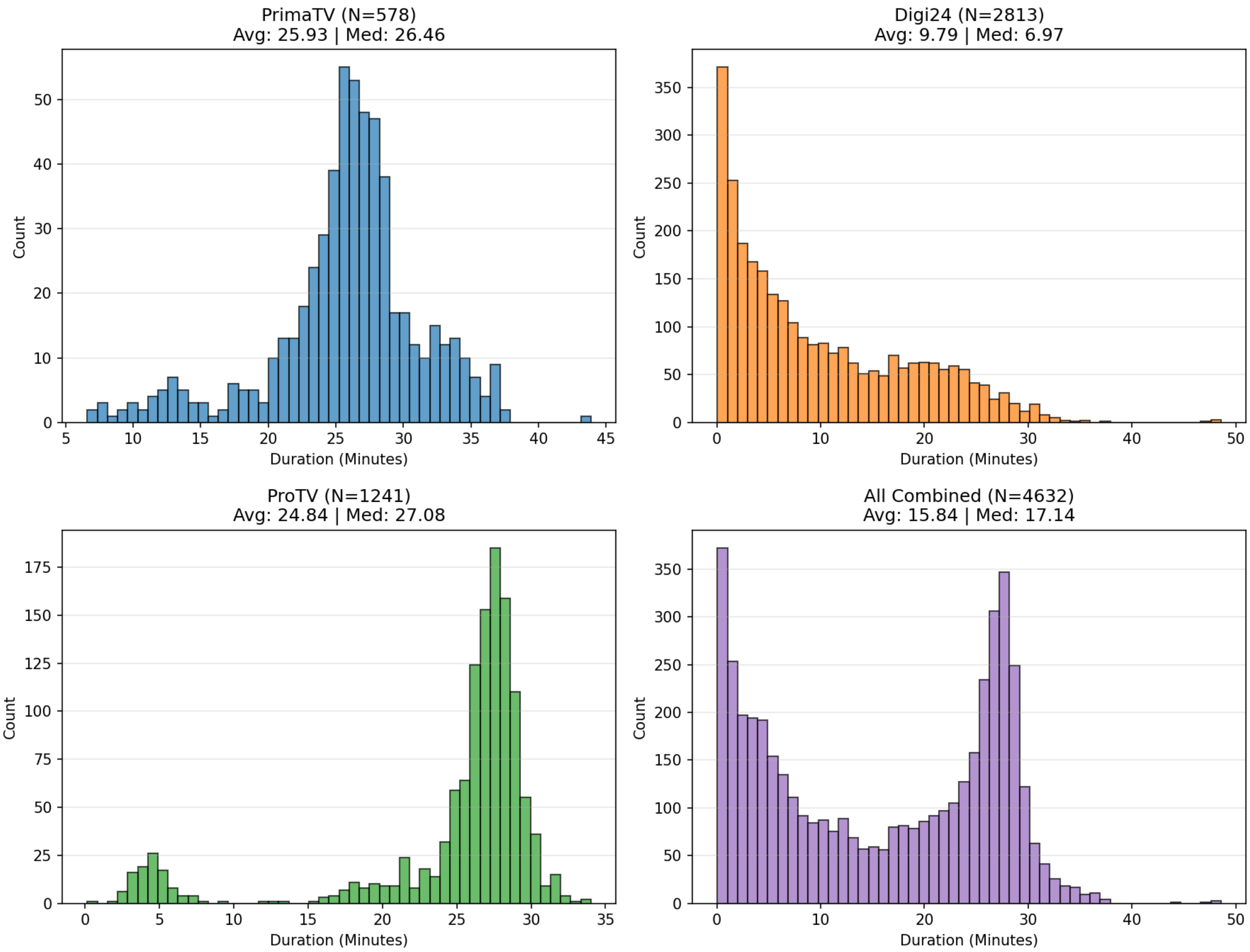

Figure 3.2: Distribution of contiguous segment durations by channel. `ProTV` and `PrimaTV` feature single long blocks; `Digi24` consists of many shorter segments.

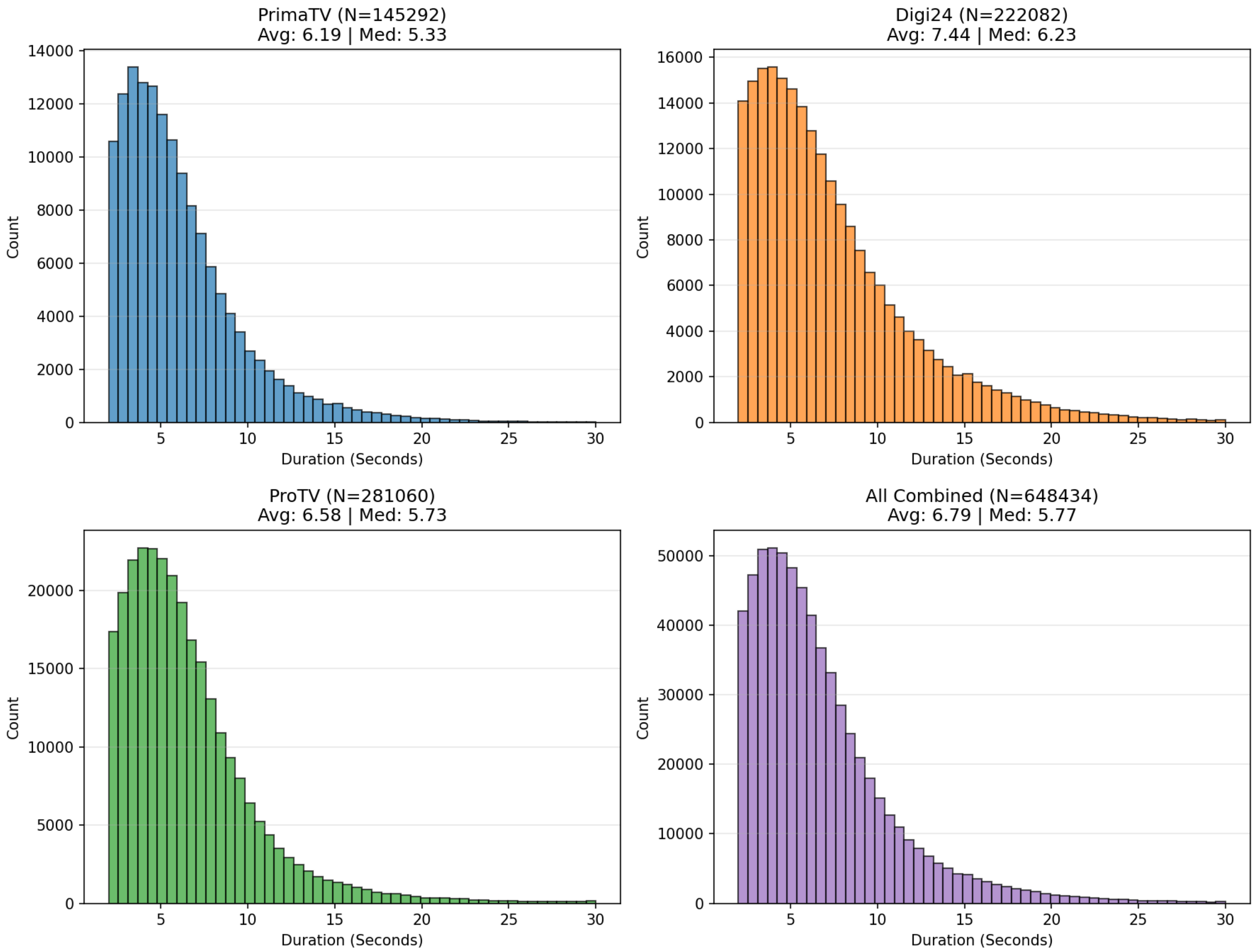

Figure 3.3: Distribution of sentence durations across the full dataset.

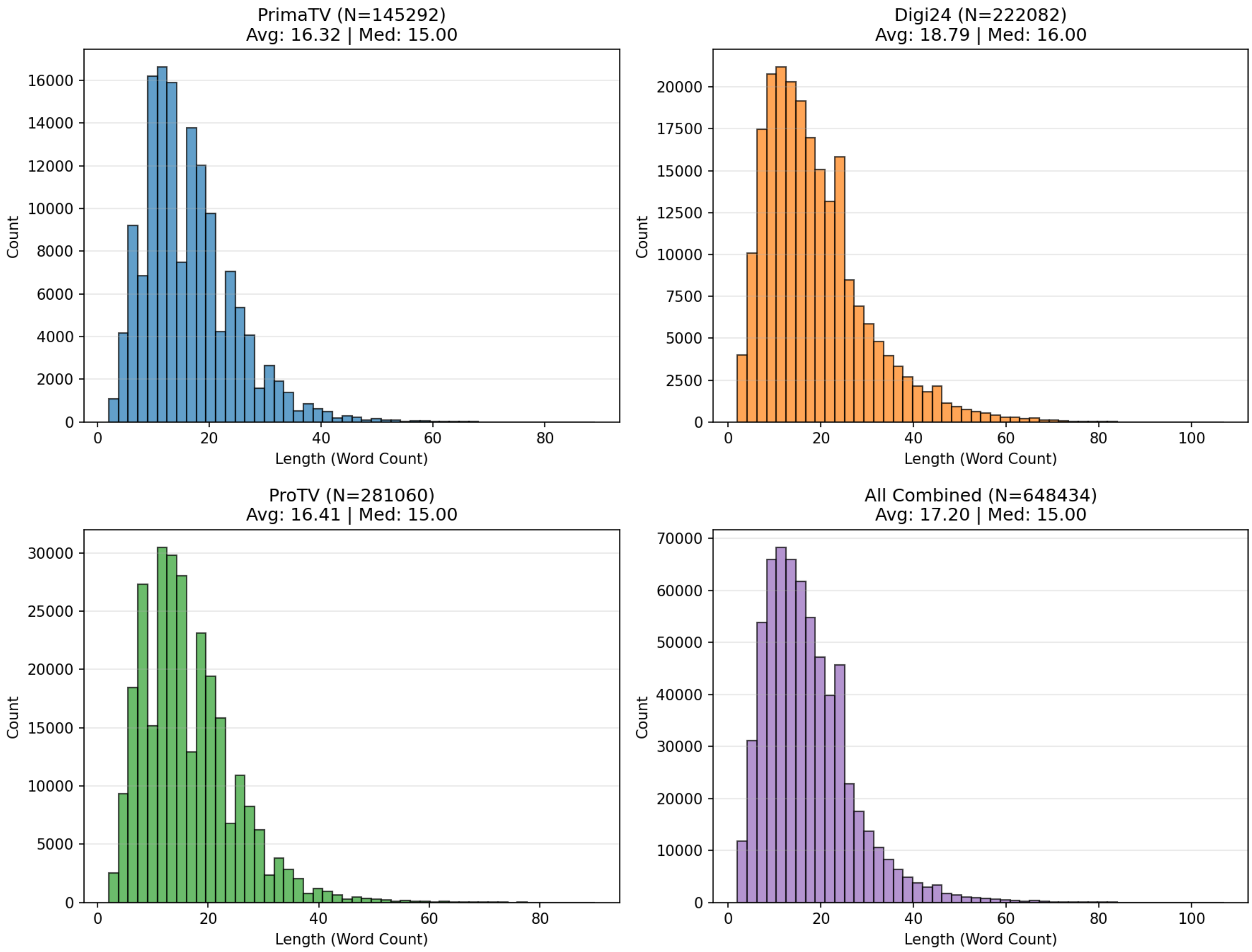

Figure 3.4: Distribution of sentence word counts across the full dataset.

3.5.1 Comparison to Existing Datasets

Table 3.3 positions `RSL-News` relative to other sign language translation datasets. The dataset overview and the best-reported scores are sourced from the recent survey by Sincan et al. [5], which independently reproduces and validates results across the field.

With 1,222 hours, the dataset ranks among the largest available, comparable in scale to `YouTube-ASL` (984h), `BOBSL` (1,467h), and `CSL-News` (1,985h). `RSL-News` has the largest vocabulary (180K) among all listed datasets, reflecting both the breadth of news-domain content and Romanian's morphological richness. By comparison, `BOBSL` achieves 77K from general BBC broadcasting, and `YouTube-ASL` reaches 60K from over 2,500 unique signers.

Unlike web-mined datasets such as `YouTube-ASL` which contains thousands of signers in uncontrolled conditions, `RSL-News` features only 6 interpreters in broadcast settings, reducing visual variability while maintaining linguistic diversity. Among diverse-domain datasets of comparable scale, `RSL-News` is the first for an under-resourced European sign language.

The best BLEU-4 column highlights the translation difficulty. `PHOENIX-2014T`, a small weather-domain dataset with only 3K vocabulary, achieves 26.75; `CSL-Daily` reaches 21.61

with 2K vocabulary. Performance degrades as vocabulary size and domain diversity increase: `How2Sign` (16K vocab) peaks at 15.5, and `BOBSL` (77K vocab, 1,467h) reaches only 7.3. Our best result of 11.16 BLEU-4 on `RSL-News` (180K vocabulary) test split is consistent with this trend and even exceeds the `BOBSL` state of the art despite a larger vocabulary.

Table 3.3: Comparison of RSL-News against existing sign language translation datasets. Best BLEU-4 scores and statistics sourced from [5].

Dataset	Language	Vocab	Hours	Signers	Source	Best BLEU-4
PHOENIX-2014T [7]	DGS	3K	11	9	TV (weather)	26.75
CSL-Daily [34]	CSL	2K	23	10	Lab	21.61
How2Sign [4]	ASL	16K	79	11	Lab	15.5
iLSU-T [35]	LSU	38K	202	18	TV (news)	3.43
OpenASL [36]	ASL	33K	288	220	Web	13.21
YouTube-ASL [6]	ASL	60K	984	>2,500	Web	–
BOBSL [12]	BSL	77K	1,467	39	TV (general)	7.3
CSL-News [8]	CSL	5K	1,985	–	TV (news)	–
RSL-News (Ours)	**RSL**	**180K**	**1,222**	**6**	**TV (news)**	**11.16**

3.6 Interpreter Analysis

Six interpreters appear across the dataset, identified by face recognition and then verified by hand. The pipeline reports seven IDs rather than six, because one of them appears on two channels: Interpreter 5 on `PrimaTV` and Interpreter 7 on `Digi24` are the same person. They are split across the channels as follows:

- **ProTV:** Interpreters 1 and 2.
- **PrimaTV:** Interpreters 3 and 5.
- **Digi24:** Interpreters 6, 7, and 8.

Figure 3.5 shows a frame from each one, and the differences are easy to see: appearance, how much space they use while signing, the background. Figure 3.6 breaks down how much of the total runtime each interpreter accounts for, and it is uneven; one main interpreter carries most of the `Digi24` and `PrimaTV` airtime, with the others stepping in only occasionally.

With several interpreters, the signing varies in style, in speed, in how far the gestures reach, and in how much the interpreter mouths. Some mouth the words clearly as they sign, so their lips closely track the spoken content; others sign with almost no mouthing. That gap is why the mouthing features turn out to help in some parts of the data much more than others, something we discuss in Chapter 4.

Figure 3.5: Representative frames showing all interpreters in the dataset.

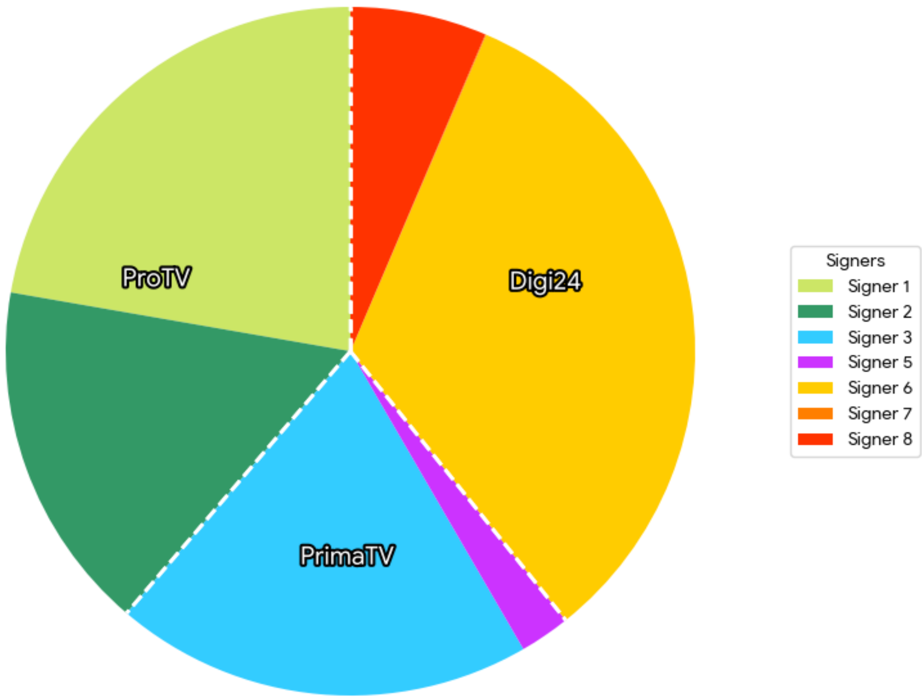

Figure 3.6: Proportion of total dataset duration by interpreter.

Chapter 4

Methodology

4.1 SignFormer Architecture and Bug Fixes

We use `SignFormer` [1] as the model of choice for all experiments. The model is a transformer encoder-decoder operating on pre-extracted I3D features (1024-d, spatially mean-pooled vectors extracted from overlapping sliding windows). The encoder uses Gloss Attention layers with 1D convolution modules, while the decoder is a standard transformer decoder with cross-attention over encoder outputs. Text targets are tokenized with the SentencePiece Unigram model trained on the lowercase RSL-News corpus. The output embedding matrix is tied to the decoder's softmax projection (`tied_softmax`) to optimize the parameter count.

4.1.1 Identified Bugs

The original `SignFormer` implementation contains two bugs in its transformer layers that prevented proper convergence on our dataset. They are both different from the standard transformer architecture that the paper describes and actively hurt the network's gradient computations.

Bug 1: Exploding Identity Mapping. The `PositionwiseFeedForward` module's forward pass returned `pwff_layer(x_norm) + x`, incorporating a residual connection internally. However, the decoder layer also adds an external identity connection via `x = x + self.feed_forward(x)`. This causes $2x + \texttt{FFN}(\texttt{LN}(x))$ to be computed instead of the correct $x + \texttt{FFN}(\texttt{LN}(x))$. This doubles the identity connection magnitude and destabilizes the gradient flow.

Bug 2: Vanishing Identity Mapping. In the encoder, residual connections are managed by a wrapper `ResidualConnectionModule`. To implement half-step residuals, the feed-forward module is initialized with a `module_factor` of 0.5. However, the original code accidentally set the identity multiplier (`input_factor`) equal to this `module_factor`, resulting in the computation $0.5 \cdot \texttt{FFN}(x) + 0.5x$. While scaling the residual $\texttt{FFN}(x)$ by 0.5 is intentional, the residual connections must preserve the identity mapping ($+1.0x$) to not interrupt the gradient flow. By scaling the identity in half, the skip connection reduced the underlying signal and gradient flow.

Fixes. We removed the internal `+x` from `PositionwiseFeedForward.forward` to prevent identity doubling in the decoder. We also explicitly set `self.input_factor = 1.0` in the `ResidualConnectionModule`, restoring the proper identity mapping.

4.1.2 Adaptation to RSL-News

Beyond the bug fixes, we made several adaptations for training on RSL-News:

- **Optimizer:** AdamW with $\beta_1 = 0.95$, $\beta_2 = 0.998$, learning rate 10^{-3}.
- **Scheduler:** Reduce the learning rate on plateau (factor 0.8, patience of 10 epochs, based on validation BLEU-4).
- **Batch size:** 100 sentences.
- **Sequence handling:** To keep memory manageable, input features exceeding 256 frames are randomly subsampled (training) or uniformly `(linspace)` subsampled (evaluation).
- **Label smoothing:** 0.1.
- **Tied softmax:** The embedding matrix is tied with the decoder's softmax projection.
- **Beam search:** Beam search is used during validation for decoding with a beam width of 4 and a length penalty of 0.6.
- **Hidden size:** 256.
- **CoPE**: No CoPE as the SignFormer ablation experiments showed that it did not improve performance on larger (256) hidden sizes, yet it doubled the parameter count.
- **Data splitting**: 90/5/5 train/val/test splits.

4.2 Decalage Modeling

4.2.1 The Alignment Problem

In live settings specific to news broadcasting, sign language interpreters translate with a temporal delay with respect to the spoken content. This delay is known as *décalage*, and means that the timestamps marking the sentence boundaries of the speech do not exactly correspond to the actual signing boundaries. When extracting a video segment between the speech start and end timestamps of a sentence, the signing for that sentence may not yet have begun at the start of the segment and may continue well beyond the end.

Figure 4.1 illustrates this phenomenon. The reference transcription is `"Parchetul de pe lângă judecătoria sectorului 1 nu a anunțat deocamdată dacă a extins ancheta."` The annotated end is at approximately 5.5 seconds (the red dashed line), but the cross-attention map shows that the model generates tokens corresponding to this sentence by attending to frames until approximately 2 seconds past the annotation boundary.

Furthermore, the first ~ 2.5 seconds contain signing from the previous sentence. The target content appears shifted by approximately 2-2.5 seconds.

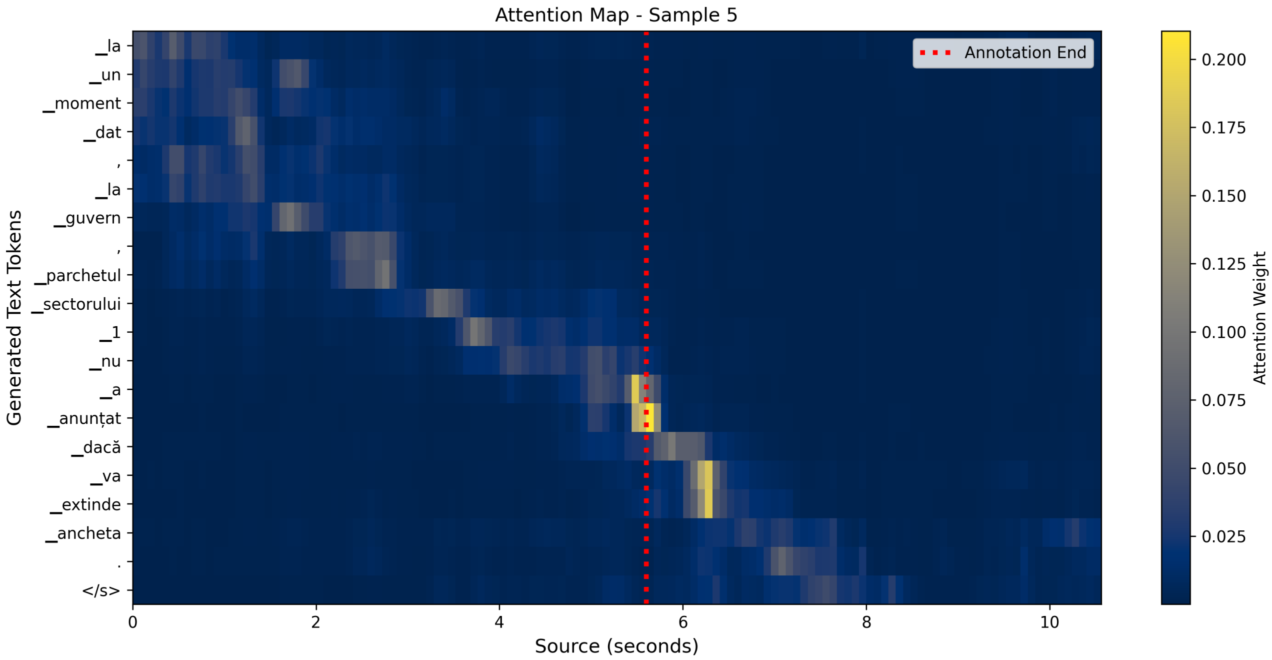

Figure 4.1: Cross-attention map showing décalage.

4.2.2 Extra Content Strategies

Since precise sign boundaries are unknown, we provide additional frames beyond the speech-derived end timestamp *(post-extension)* and let the attention mechanism learn which frames to attend to. At evaluation time, we used a fixed 4-second post-extension with uniform subsampling for determinism. During training, we explore several strategies:

1. **Random (2, 4) seconds:** Uniform random post-extension between 2 and 4 seconds.
2. **Random (0, 5) seconds:** Wider range, 0 to 5 seconds.
3. **Fixed 5 seconds:** Deterministic 5-second extension at training time.
4. **Random (1, 5) seconds:** Final range used in the best configuration.

Wider random ranges are more likely to catch the décalage and also act as data augmentation; the model cannot memorize fixed boundary positions and must learn to attend to relevant content within windows of various sizes. The results are presented in Chapter 5.

4.2.3 Confidence Ramps

In order to model uncertainty near boundaries, we introduce *confidence ramps*, which scale the feature vectors based on the confidence of each frame. The ramp is a per-frame scalar which is multiplied element-wise with the feature vectors, scaling down frames where we are less confident they belong to the current sentence.

Start ramp (fade-in). Frames at the beginning of a segment may contain leftover signing from the previous sentence. We compute a maximum boundary window based on the temporal gap to the previous sentence's end timestamp:

$$\texttt{ramp\_max} = \texttt{clip}(1.5 - 0.26 \times \texttt{gap\_seconds},\ 0.2,\ 1.5)$$

A short gap (sentences close together) produces a larger boundary window, reflecting a higher probability of boundary overflow. During training, the actual duration of this boundary window is uniformly sampled up to `ramp_max`. Within this window, the fade-in scales linearly from 0 to 1 over *at least half* of the frames, while the remaining frames are fully confident. Specifically, the fully confident fraction of the window is uniformly sampled between 0 and 0.5 during training, and fixed at 0.25 (with the window duration halved) during evaluation.

Post-extension ramp (fade-out). After the annotated end timestamp, the post-extension window is added. The total duration of this extension is uniformly sampled during training and fixed at 4.0 seconds during evaluation. Within this extension, frames initially remain at full confidence (as the interpreter is likely still signing), then decay linearly to 0 toward the end of the window. The fraction of fully confident frames within the extension is randomized during training (uniform between 0.3 and 0.7) and fixed at 0.5 during evaluation.

The ramp is applied multiplicatively to both sign features and phoneme logits before they enter the encoder.

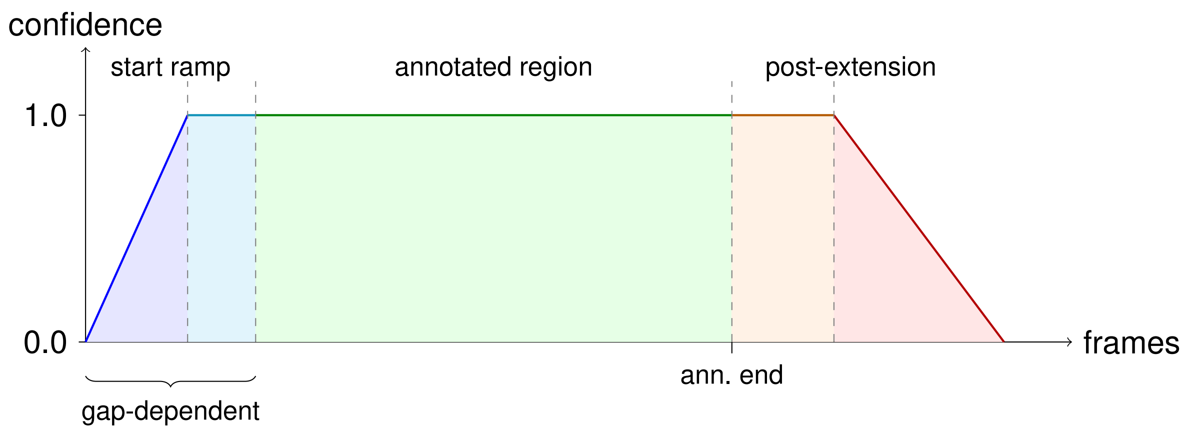

Figure 4.2: Confidence ramp structure.

4.3 Mouthing Feature Extraction

Sign language interpreters usually mouth the words they are signing. These mouthings provide additional phonemic information that can disambiguate otherwise similar manual signs. To introduce these phoneme-level features, we fine-tune a visual speech recognition model on videos of mouth regions and corresponding phoneme labels.

4.3.1 Data Preparation

We used an existing 100+ hours Romanian Visual Speech Recognition dataset as training data. To prepare it, we perform the following:

1. **Dataset cleaning:** Filtering and normalizing the `ro_vsr` [45] dataset.
2. **Phonemization:** Converting Romanian text labels to sequences of phonemes using the `RoLEX` lexicon [33].
3. **OOV handling:** Words not found in `RoLEX` are phonemized using an LLM-based fallback.

The final label set consists of 37 phonemes plus a CTC blank token and a shared sos/eos token (39 classes total). The sos/eos token is later dropped from the output logits used as features, giving us 38-dimensional vectors per frame.

4.3.2 Auto-AVSR Fine-tuning

We fine-tune `Auto-AVSR` [28] in video-only mode. The architecture consists of a ResNet-18 visual frontend, a 12-layer Conformer encoder, and a hybrid CTC/attention decoder.

Frozen layers. The frontend and the first 6 encoder layers are loaded from the pretrained English checkpoint and frozen. The remaining 6 encoder layers are randomly initialized and trainable, along with the CTC head and attention decoder.

Loss configuration. Because signer mouthings differ fundamentally from natural speech (they are sporadic, partial, and asynchronous with the manual signs), the fine-tuning requires a specific CTC/attention configuration:

- **Higher attention weights** (e.g., 50% or more) cause the model to hallucinate complete words not actually mouthed because it over-fits to natural speech patterns where words appear in full.
- **CTC only** produces temporally scrambled outputs when applied to mouthing because it lacks any sequence-level regularization.
- **90% CTC / 10% attention** (our chosen configuration) yields 29.23% PER on natural speech. The CTC component provides strong frame-level alignment, while minimal attention acts as a light regularizer, giving just enough language understanding to handle mouthings without hallucinating.

Training details. The learning rate is 10^{-3} for new layers, 3×10^{-4} for the unfrozen encoder layers. Training was performed for a maximum of 50 epochs with a 5-epoch warmup and cosine annealing.

4.3.3 Feature Generation

The fine-tuned model is applied to the mouth region crops from all RSL-News interpreter videos. For each segment, we extract per-frame CTC logits (38-dimensional after dropping the sos/eos token), storing them as compressed `.npz` files alongside the sign features. At training time, the phoneme logits are resampled via uniform indexing to match the sign feature sequence length (since the visual speech model and `I3D` operate at different temporal resolutions).

4.3.4 Analysis of Mouthing Predictions

To illustrate what the fine-tuned model captures from signer mouthings, we present an example that has the following spoken sentence corresponding to the signed segment:

Drumul de centură al oraşului Călimăneşti din Vâlcea este în continuare închis.

The model's CTC output (greedy decoding, X-SAMPA notation) is:

`n u m i n e S t i t u n o r a S a l e b e l e t f i gZ e z o d e l s`

The corresponding viseme sequence (grouping phonemes by visual mouth shape) is:

`t/d/n u p/b/m i t/d/n e S/tS/Z/gZ t/d/n i t/d/n u t/d/n o r a S/tS/Z/gZ a l e p/b/m e l e t/d/n f/v i S/tS/Z/gZ e s/z/ts o t/d/n e l s/z/ts`

We can observe that the model successfully identifies the core content words that the interpreter actually mouths, while skipping the grammatical function words (*de*, *al*, *din*, *este*, *în*) that are typically not mouthed during signing. This aligns with sign language grammar, which omits function words and means the phoneme features do not inject irrelevant noise.

Second, the predictions are consistent with the limitations of the visual medium for speech recognition. Lip-reading operates on *visemes* (visually distinguishable mouth configurations), and not phonemes. Multiple phonemes map to the same viseme: voiced/unvoiced pairs like /d/-/t/-/n/ share identical lip and tongue positions, as do /f/-/v/, /s/-/z/, and /S/-/Z/. Sounds produced in the back of the mouth or by the vocal cords are invisible on the lips. This means that the model's "errors" are often not errors at all but inherent ambiguities:

- **"Drum"** → `n u m`: /d/ and /n/ share the same viseme; /r/ is formed at the back of the mouth and swallowed by the lip-rounding of /u/. Visually, "drum" and "num" are indistinguishable.
- **"Călimăneşti"** → `n e S t i`: the model captures the salient back-half of the word. /S/, /t/, and /i/ produce highly distinct lip configurations.
- **"Oraş"** → `o r a S`: a perfect phonetic match for the word root.
- **"Vâlcea"** → `f i gZ e`: /v/ and /f/ share the same viseme; /1/ *(â)* looks like /i/; /tS/ (the *ch* in "Vâlcea") and /gZ/ share the same puckering of the lips.

The tail of the output (`z o d e l s`, which should have corresponded to "continuare închis") is noise. This might be explained by the interpreter having stopped mouthing or already having transitioned to the next sentence. This is consistent with the nature of signer mouthings, as discussed earlier.

While these are not phonetically perfect predictions, these features will help the translation model disambiguate signs of visually similar concepts (e.g., "city" vs. "town") or proper nouns like city names, which are often fingerspelled too fast for recognition.

4.4 Mouthing Feature Integration

4.4.1 Gate Collapse

Our initial fusion design used a sigmoid gate that chose between the sign and phoneme feature streams:

$$F_G = \sigma(W_g \cdot \mathbf{sgn}), \quad \mathbf{h} = F_G \odot \mathbf{sgn} + (1 - F_G) \odot \mathbf{phn},$$

where $W_g \in \mathbb{R}^{D \times D}$ was a full matrix producing a per-dimension gate from the sign embeddings. However, at approximately 360k training steps, the gate collapsed. Because sigmoid is steep around 0, small parameter changes caused the gate to suddenly saturate toward the phoneme stream. Since the phoneme features are noisy and only weakly correlated with the target natural sentences, the translation quality dropped from 8.72 BLEU-4 to 6.31 and could never recover.

Replacing the matrix gate with a scalar gate (per-frame, per-dimension-shared) prevented the collapse but introduced a different problem: the scalar gate did not allow enough of the sign signal to flow through, effectively bottlenecking the primary information source.

4.4.2 Additive Residual Fusion

The solution we came up with is to always preserve the full sign stream and treat phoneme information as a gated (weighted) residual:

$$\alpha = \sigma(W_g \cdot \mathbf{sgn} + b_g), \quad \mathbf{h} = \mathbf{sgn} + \alpha \cdot \mathsf{LN}(\mathsf{proj}(\mathbf{phn})),$$

where $W_g \in \mathbb{R}^D$ outputs a single scalar per frame, proj is a linear projection from the phoneme dimension (38) to the model's embedding dimension D, and LN is layer normalization. The gate bias b_g is initialized to -2.0, so $\sigma(-2) \approx 0.12$ at the start of training (so the model initially relies almost entirely on sign features and gradually learns when and how much to incorporate mouthing information).

The fusion module is placed after the sign embedding layer and before the encoder, so both sign and phoneme information are processed together by the self-attention layers. The confidence ramps described in Section 4.2.3 are applied to both before fusion.

4.5 Use of AI

`Gemini` [46] was used with the purpose of rephrasing a few certain paragraphs that we found to be unclear or verbose. The AI was instructed to maintain the technical content and meaning of the original text and to only improve the flow of the writing. We reviewed all AI-generated suggestions and made final edits to ensure that the changes were consistent with the rest of the document. `Claude` [47] was used for the purpose of writing a few certain snippets of boilerplate and repetitive code as part of the dataset cleaning pipeline.

Chapter 5

Experiments and Results

5.1 Metrics

BLEU (Bilingual Evaluation Understudy) [48] measures translation quality by computing precision over n-grams (consecutive word sequences of length 1 to N) between the generated hypothesis and a reference translation. BLEU-N is the geometric mean of n-gram precisions for $n = 1, \ldots, N$, multiplied by a brevity penalty that discourages overly short outputs.

PER (Phoneme Error Rate) measures the edit distance (insertions, deletions, substitutions) between a predicted phoneme sequence and the reference, normalized by the reference length. We use PER to evaluate the mouthing extraction model on natural Romanian speech.

5.2 Experimental Setup

All experiments in the following comparisons use the validation split of the RSL-News dataset. The data is split 90/5/5 (train/val/test) with a fixed seed of 42. We report BLEU-4 on the validation split using beam search (size 4, length penalty $\alpha = 0.6$). Evaluation always uses a fixed 4-second post-extension and deterministic uniform (linspace) subsampling.

Training is performed on a single NVIDIA RTX 4070 Ti (12 GB VRAM). Each epoch takes approximately 8 minutes including validation. The only variables between the experiments are the bug fixes, extra content strategy, ramp configuration, and phoneme integration.

5.3 Consolidated Results

Before going into the validation experiments in the upcoming sections, we first present the final results on the test split. Once the best configuration was established (Experiment 11), we ran a grid search on the validation split to select the optimal beam search parameters (size 10, $\alpha = 1.0$).

Evaluating this final setup on the test split achieved an overall BLEU-4 of **11.16**. The breakdown by news channel is **12.92** for **Digi24**, 11.16 for **ProTV**, and 8.28 for **PrimaTV**.

The lower score for PrimaTV is likely due to the fact that there is less training data available for this channel, combined with its lower video resolution, which probably resulted in lower-quality features.

Table 5.1 presents all experimental configurations in chronological order, along with the improvements each contributed with.

Table 5.1: Progressive results across all configurations. “Rand (a, b)” means random post-extension between a and b seconds during training.

#	Configuration	BLEU-4	Epochs
1	Original SignFormer (both bugs present)	2.74	80
2	First bug fix only	2.86	82
3	First fix + rand (2, 4)s extra	3.57	97
4	Both fixes + rand (2, 4)s extra	6.56	103
5	Both fixes + rand (0, 5)s extra	9.19	209
6	Both fixes + flat 5s extra	9.31	205
7	Both fixes + rand (2, 4)s + ramps	8.95	61
8	Both fixes + rand (1, 5)s + ramps	9.98	188
9	Phonemes + flat 5s extra	10.01	199
10	Phonemes + flat 5s + 1 epoch ramps	10.30	200
11	**Phonemes + rand (1, 5)s + ramps**	**10.76**	200

Experiment 10 was trained for 199 epochs without ramps (achieving 10.01 BLEU-4 when evaluated without ramps, and 8.78 BLEU-4 when evaluated with ramps), then trained for one additional epoch with ramps enabled and evaluated with ramps, jumping to 10.30 BLEU-4. This demonstrates that ramps provide an immediate benefit even when introduced late in training. Experiment 11, trained from scratch with all components, achieves the best overall result of 10.76 BLEU-4.

5.4 Ablation Analysis

We isolate the contribution of each component by comparing configurations that differ in exactly one variable.

Bug fixes: +3.82 BLEU. Comparing experiments 3 and 4 (the same extra content strategy, only difference is the second bug fix): 3.57 $\rightarrow$ 6.56. The first fix alone contributes only +0.12 (exp. 1 $\rightarrow$ 2), but together the improvement is already notable.

Extra content range: +2.75 BLEU. Comparing experiments 4 and 6 (both bugs fixed, narrow random vs. fixed wide extra): 6.56 $\rightarrow$ 9.31. Wider post-extension captures more of the delayed signing. The random (0, 5) variant (9.19) performs comparably to the fixed 5s (9.31), indicating that the actual amount of extra content matters more than whether it is randomized.

Confidence ramps (same range): +2.39 BLEU. Comparing experiments 4 and 7 (rand(2, 4) without vs. with ramps): 6.56 $\rightarrow$ 8.95. Ramps provide a large benefit even with a narrow extra range, because they explicitly model boundary uncertainty. With a wider range (experiment 6 vs. 8): 9.31 $\rightarrow$ 9.98, the improvement is smaller (+0.67) because the wide extra content already implicitly handles some boundary uncertainty.

Phoneme features: +0.78 BLEU. Comparing experiments 8 and 11 (best non-phoneme vs. best with phonemes, same strategy): 9.98 $\rightarrow$ 10.76. The mouthing features provide complementary information for the attention mechanism. The improvement is on top of the ramps.

Combined effect. From the broken baseline (2.74) to the best configuration (10.76): a total improvement of +8.02 BLEU-4.

5.5 Comparative Analysis

Table 5.2: Comparison of SLT results on various datasets.

Dataset	Language	Hours	Vocab	Domain	BLEU-4
CSL-Daily	CSL	23	2K	Controlled (daily)	21.61
How2Sign	ASL	79	16K	Controlled (instr.)	15.5
PHOENIX-2014T	DGS	11	3K	Wild (weather)	26.75
iLSU-T	LSU	202	38K	Wild (news)	3.43
BOBSL	BSL	1,467	77K	Wild (diverse)	7.3
RSL-News	**RSL**	**1,222**	**180K**	**Wild (news)**	**11.16**

Table 5.2 shows that RSL-News achieves the highest BLEU-4 among all diverse-domain datasets despite having the largest vocabulary (180K). This result validates both the dataset quality and the effectiveness of our modeling approach. This comparison also shows how difficult the wild setting and diverse-domain are compared to the controlled, narrow-domain ones: even with 1,222 hours of data, wild news-domain translation (11.16) scores far below single-domain benchmarks (26.75 on weather).

Relative to the most comparable system (iLSU-T: 202 hours of news-domain broadcast SLT achieving 3.43 BLEU-4), our pipeline achieves a 3$\times$ improvement.

5.6 Qualitative Analysis

The following are translation examples from the validation set chosen at random, illustrating interesting aspects of the system.

5.6.1 Semantic Paraphrasing

Reference: *intervenția a fost dificilă, pentru că victima era prinsă în interiorul autoturismului.*

Hypothesis: *intervenția a fost dificilă pentru că victima era încarcerată în caroseria distrusă.*

The model captures the core meaning correctly (difficult intervention because the victim was trapped in the vehicle) but paraphrases: "încarcerată în caroseria distrusă" (incarcerated in the destroyed car body) instead of "prinsă în interiorul autoturismului" (caught inside the car). This is expected as the interpreter signs the *concept*, not the exact words, and the model translates what it sees.

5.6.2 Compression

Reference: *în timpul unei discuții a prins bărbatul şi a ieşit din minţe, a pus mâna pe un cuțit şi a omorât-o pe femeia de 28 de ani.*
Hypothesis: *bărbatul a luat cuțitul şi a ucis-o pe femeie în vârstă de 28 de ani.*

The model produces a compressed but semantically accurate translation. The context ("during a discussion, he lost his mind") is dropped, but the core event (man took knife, killed 28-year-old woman) is preserved. Interpreters routinely compress information under time pressure, and the model has learned this pattern. The hypothesis also contains some speech recognition errors, but due to the dataset's large scale, the model learns to be robust to them.

5.6.3 Named Entity Errors

Reference: *este vorba despre favipiravir, un antiviral care poate fi administrat în primele zile de boală pentru a preveni cazurile complicate şi spitalizarea.*
Hypothesis: *este vorba despre fentanilul, un medicament care poate fi prima zi de boală pentru a preveni situația complicată şi internarea în spital.*

The sentence structure is well-preserved, and the general meaning (a drug to prevent complications and hospitalization) is correct. However, the specific drug name is wrong: "favipiravir" becomes "fentanilul." Named entities, especially technical terms fingerspelled or mouthed partially, remain a weakness. The model substitutes a phonetically plausible but incorrect medical term. The action is also missed: "administrat" (administered), possibly due to it being obvious from context and not being explicitly signed.

5.6.4 Meaning Drift

Reference: *pentru că, repet, situația este una dramatică din punct de vedere demografic.*
Hypothesis: *pentru că situația este una dramatică din punctul meu de vedere.*

The model drops "repet" (I repeat) and transforms "din punct de vedere demografic" (from a demographic point of view) into "din punctul meu de vedere" (from my point of view). The structural template is preserved, but the qualifying adjective “demografic” is lost, changing the meaning. This type of error likely stems from the interpreter simplifying or abbreviating the qualifying phrase. The generated phrase is a lot more common, so maybe the model generalized to that more common pattern.

Chapter 6

Conclusions and Future Work

This work aims to bring Sign Language Translation to Romanian, a language absent from the field of SLT, without the gloss annotations that have historically bottlenecked progress. The result is `RSL-News`, a 1,222-hour gloss-free dataset mined from six years of interpreted news programs, and the first set of SLT baselines trained on it. We also turned a compact translation model into a system that reaches 11.16 BLEU-4 on held-out data, the highest score reported on any broadcast-domain sign language dataset so far.

One of the most interesting findings is that two small bugs in the baseline's residual connections were silently capping its accuracy, and fixing them alone produced a huge jump in accuracy. It turns out that on a new dataset, the model itself deserves as much scrutiny as the data does. The second is that décalage is not a nuisance to be ignored but a dominant difficulty in interpreted broadcast data; feeding the model a few extra seconds of video past the annotated sentence end, and then teaching it to distrust the frames near those uncertain boundaries through confidence ramps, carried it from 6.56 to nearly 9 BLEU-4 on the validation split. The third is that mouthing carries a real, if modest, signal: phoneme features read from the interpreter's lips improved translation on top of everything else, even though they are noisy and only some interpreters mouth consistently. The distance from the broken starting point to the final model is therefore explained almost entirely by adaptations specific to the interpreted-broadcast setting, not by changes to the core architecture.

Every experiment ran on a single consumer GPU, each epoch taking approximately eight minutes, the model of choice having only about 6.6 million parameters. Even so, it sits at the top of the broadcast-domain results. This is encouraging for an under-resourced language, as it places the real bottleneck in data and engineering, both reproducible, rather than in access to large-scale compute.

6.1 Future Work

Stronger visual features. The current system reads from `I3D` features pretrained on British Sign Language. In the future we could compare against, or combine with, self-supervised visual backbones such as `DINOv2` [18], which are not tied to a specific sign language and may transfer better while providing richer representations.

Better décalage modeling. Confidence ramps handle boundary uncertainty with a fixed, hand-designed shape. The lag could instead be estimated per sentence using smarter, data-driven methods.

LLM decoders. We deliberately kept the model small. Replacing the decoder with a pretrained Romanian or multilingual language model, in the spirit of `Sign2GPT` [15] and `Uni-Sign` [8], could inject the linguistic priors that a few million-parameter decoder cannot learn from the corpus alone, and is the most likely route to closing the gap with narrow-domain scores.

Mouthing supervision from pseudo-glosses. Our lip-reading model is trained on full natural speech and then applied to the sparse, partial mouthings of interpreters. This causes a mismatch we try to mitigate by giving a very low attention weight to keep the model from hallucinating whole words. A cleaner approach would be to derive pseudo-glosses for each sentence with an LLM given a set of rules defined by an expert RSL linguist. Then map them to the phonemes an interpreter is actually likely to mouth, and fine-tune the lip-reading model on that target. With a label that matches what is really on the lips, the attention weight could be raised again, and the resulting phoneme features should give a stronger signal to the translation model.

None of these directions requires new data; `RSL-News` is already large enough to support them.

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STATEMENT REGARDING
THE AUTHENTICITY OF THE THESIS PAPER

I, the undersigned Radu-Cătălin MICEA, identifying myself with CI series TM no. 1017667, CNP 5010728020078, author of the thesis paper *RSL-News: Large-Scale Dataset and Enhanced Lightweight Transformer Model for Gloss-Free Romanian Sign Language Translation,* developed with the purpose of participating in the graduation examination completing the educational level of Master's degree organized by the Faculty of Automation and Computers within the Politehnica University of Timişoara, session July of the academic year 2026, coordonated by Prof. Dr. Habil. Eng. Călin-Adrian POPA, considering Article 34 of the *Regulation on the organization and conduct of bachelor/diploma and dissertation examinations*, approved by Senate Decision no. 109/14.05.2020, and knowing that in the event of subsequent finding of false statements, I will bear the administrative sanction provided by Art. 146 of Law no. 1/2011 – law of national education, namely the cancellation of the diploma of studies, I declare on my own responsibility that:

- This paper is the result of my own intellectual endeavour,
- He paper does not contain texts, data or graphic elements taken from other papers or from other sources without such authors or sources being quoted, including when the source is another paper / other works of my own;
- bibliographic sources have been used in compliance with Romanian legislation and international copyright conventions.
- this paper has not been publicly presented, published or presented before another the bachelor/diploma/dissertation examination committee.
- In the development of the paper I have used instruments specific for artificial intelligence (AI), namely *Gemini 3.1 Pro* (https://gemini.google.com/) and *Claude Opus 4.6* (https://claude.ai/), which I have quoted within the paper.

I declare that I agree that the paper should be verified by any legal means in order to confirm its originality, and I consent to the introduction of its content in a database for this purpose.

Timişoara,
Date
28.06.2026

Signature
Micea